PLUMBING (Continued)

SYSTEM LOW

(1) Position the vehicle in a wind-free work area. This will aid in detecting small leaks.

(2) Bring the refrigerant system up to operating temperature and pressure. This is done by allowing the engine to run at idle under the following conditions for five minutes, then turning the engine off.

(a) Front windows open.

(b) Transmision in Park or Neutral with the park brake set.

(c) A/C-heater controls set to fresh air, panel mode, high blower speed and A/C compressor engaged.

(3) Wait five to seven minutes and, with the engine not running, use a electronic R-134a leak detector and search for leaks. Because R-134a refrigerant is heavier than air, the leak detector probe should be moved slowly along the bottom side of all refrigerant lines, connector fittings and components.

(4) To inspect an A/C evaporator for leaks, insert the electronic leak detector probe into a floor outlet or the recirculation air intake.

STANDARD PROCEDURE

REFRIGERANT SYSTEM SERVICE EQUIPMENT

WARNING: Refer to the applicable warnings and cautions for this system before performing the following operation (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - WARNINGS) and (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - CAUTIONS). Failure to follow the warnings and cautions could result in possible personal injury or death.

WARNING: Eye protection must be worn when servicing an air conditioning refrigerant system. Turn off (rotate clockwise) all valves on the equipment being used, before connecting to or disconnecting from the refrigerant system. Failure to observe these warnings may result in personal injury or death.

RECOVERY/RECYCLING STATION

When servicing the A/C system, a R-134a refrigerant recovery/recycling/charging station that meets SAE Standard J2210 must be used (Fig. 1). Contact an automotive service equipment supplier for refrigerant recovery/recycling/charging equipment. See the operating instructions supplied by the equipment manufacturer for proper care and use of this equipment.

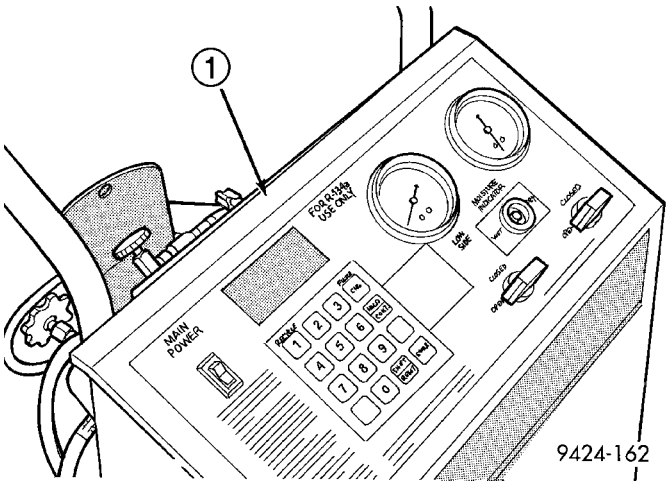

Fig. 1 Refrigerant Recovery/Recycling Station - Typical

1 - R-134a REFRIGERANT STATION

MANIFOLD GAUGE SET

A manifold gauge set may be needed with some recovery/recycling/charging equipment (Fig. 2). The service hoses on the gauge set being used should have manual (turn wheel), or automatic back-flow valves at the service port connector ends. This will prevent refrigerant from being released into the atmosphere.

MANIFOLD GAUGE SET CONNECTIONS

CAUTION: Do not use an R-12 manifold gauge set on an R-134a system. The refrigerants are not compatible and system damage will result.

- **LOW PRESSURE GAUGE HOSE** - The low pressure hose (Blue with Black stripe) attaches to the suction (low-side) service port. This port is located on the A/C liquid line.
- **HIGH PRESSURE GAUGE HOSE** - The high pressure hose (Red with Black stripe) attaches to the discharge (high-side) service port. This port is located on the A/C discharge line.
- **RECOVERY, RECYCLING, EVACUATION AND CHARGING HOSE** - The center manifold hose (Yellow, or White, with Black stripe) is used to recover, evacuate, and charge the refrigerant system. When the low or high pressure valves on the manifold gauge set are opened, the refrigerant in the system will escape through this hose.